

LSQ 实验报告



代码:

```
#include <stdio.h>
#include <math.h>

#define m 2

double* gauss(double a[m + 1][m + 1], double b[m + 1], double x[m + 1]) {

    double l[m + 1][m + 1];
    double sum;
    int i, j, k;

    for (k = 0; k <= m - 1; k++) {
        for (i = k + 1; i <= m; i++) {
            l[i][k] = a[i][k] / a[k][k];
            b[i] = b[i] - l[i][k] * b[k];

            for (j = k + 1; j <= m; j++) {
                a[i][j] = a[i][j] - l[i][k] * a[k][j];
            }
        }
    }

    x[m] = b[m] / a[m][m];

    for (k = m - 1; k >= 0; k--) {
        sum = 0;
        for (j = k + 1; j <= m; j++) {
            sum = sum + a[k][j] * x[j];
        }
        x[k] = (b[k] - sum) / a[k][k];
    }

    return x;
}

int main() {
    double A[3][3] = {{0, 0, 0}, {0, 0, 0}, {0, 0, 0}};
    double x[9] = {0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9};
    double y[9] = {5.1234, 5.3057, 5.5680, 5.9378, 6.4370, 7.0978, 7.9493,
, 9.0253, 10.3627};
}
```

```

double B[3] = {0, 0, 0};
double F[3] = {0, 0, 0};

for (int i = 0; i < 9; i++) {
    A[0][0] += 1;
    A[0][1] += x[i];
    A[0][2] += pow(x[i], 2);
    A[1][2] += pow(x[i], 3);
    A[2][2] += pow(x[i], 4);

    A[1][0] = A[0][1];
    A[1][1] = A[0][2];
    A[2][0] = A[0][2];
    A[2][1] = A[1][2];
}

for (int j = 0; j < 9; j++) {
    B[0] += y[j] * 1;
    B[1] += y[j] * x[j];
    B[2] += y[j] * pow(x[j], 2);
}

printf("正则方程组系数矩阵:\n");
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        printf("%lf ", A[i][j]);
    }
    printf("\n");
}

printf("右端项矩阵:\n");
for (int i = 0; i < 3; i++) {
    printf("%lf ", B[i]);
    printf("\n");
}

double* result = gauss(A, B, F);

printf("解向量:\n");
for (int i = 0; i < 3; i++) {
    printf("%lf ", result[i]);
}

printf("\n");
printf("拟合多项式为: \n");
printf("y=%lf+%lfx+%lfx^2", result[0], result[1], result[2]);

```

```
    return 0;  
}
```

运行结果:

```
正则方程组系数矩阵:  
9.000000 4.500000 2.850000  
4.500000 2.850000 2.025000  
2.850000 2.025000 1.533300  
右端项矩阵:  
62.807000  
35.207360  
23.944224  
解向量:  
5.305581 -1.825190 8.164957  
拟合多项式为:  
y=5.305581+-1.825190x+8.164957x^2  
-----  
Process exited after 0.08176 seconds with return value 0  
请按任意键继续. . .
```